



COMPETENCY STANDARD
FOR
PLUMBING AND PIPE FITTING
(CONSTRUCTION SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Plumbing and Pipe Fitting** is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

The document was developed under the Skills for Employment Investment Program (SEIP) and subsequently reviewed and updated with the engagement of occupation-specific industry experts to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level-15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: <https://sicip.gov.bd/>

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Occupation Specific Competencies (120 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-CON-PPF-1-O	Fabricate and Prepare Pipes for Installation	1. Practice OSH 2. Collect tools, equipment and materials 3. Carry out pipe cutting operation 4. Perform threading 5. Clean and maintain the work area, tools & equipment.	15
SICIP-CON-PPF-2-O	Make Pipe Joints	1. Interpret drawings and plumbing plans 2. Cut wall and floor 3. Assemble and Join Pipes and Fittings 4. Lay pipes with fittings 5. Clean and maintain the work area, tools & equipment	15
SICIP-CON-PPF-3-O	Install Water Supply and Wastewater Pipeline	1. Gather and inspect tools, equipment and materials 2. Fix joints and fittings for pipe 3. Install water supply pipes 4. Install wastewater pipes 5. Clean and maintain the work area, tools & equipment	40
SICIP-CON-PPF-4-O	Carry Out Plumbing Fixtures Installation and Testing	1. Prepare for plumbing fixture installation 2. Gather tools, equipment and materials 3. Install plumbing fixtures 4. Perform leak and pressure testing 5. Clean and maintain the work area, tools & equipment	40
SICIP-CON-PPF-5-O	Perform Installation of Hot Water Piping System.	1. Prepare for works 2. Install pipes on the wall surface 3. Install geyser with service line 4. Clean and maintain the work area, tools & equipment	10
Total Hours			120

The Occupation Specific Competencies

Unit of Competency: FABRICATE AND PREPARE PIPES FOR INSTALLATION	Nominal Duration: 15 Hrs.	Unit Code: SICIP-CON-PLF-1-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to fabricate and prepare pipes for installation. It specifically includes the tasks of practicing OSH, collecting tools, equipment and materials, carrying out pipe cutting operation, performing threading, cleaning and maintaining the work area, tools & equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Practice OSH	1.1 OHS policies and safe operating procedures are understood. 1.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 1.3 First aid procedure for emergency situations is understood.
2. Collect tools, equipment and materials	2.1 Appropriate <u>hand tools</u> and <u>power tools</u> are selected and prepared as per job requirement. 2.2 <u>Tools, equipment and materials</u> are identified to do the job. 2.3 Proper and safe use/operation is applied in the different types of hand tools. 2.4 Power tools are used safely in accordance to manufacturer's operating specification.
3. Carry out pipe cutting operation	3.1 <u>Pipes</u> are clamped and fixed using pipe vice. 3.2 Pipe is measured and marked as per drawing. 3.3 Pipes are cut using appropriate tools. 3.4 Newly cut pipe is deburred and removed of burr using appropriate <u>deburring tool</u> .
4. Perform threading	4.1 Pipes are clumped with pipe vice. 4.2 Diestocks are adjusted and initiated into the pipe end squarely. 4.3 Thread cutting is carried out on pipe in accordance to workplace procedure. 4.4 <u>Coolant</u> is used during thread cutting operation. 4.5 Threads are checked to conform with the specified form and measurement in accordance with plan/drawing.

	4.6 Newly threaded pipe is deburred and removed of burr using appropriate tools.
5. Clean and maintain the work area, tools & equipment	5.1 Plumbing tools and equipment are cleaned and stored in designated area. 5.2 Work area is cleaned and waste materials are disposed in accordance with workplace requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. PPE	1.1 Apron 1.2 Safety helmet 1.3 Safety shoes 1.4 Hand gloves 1.5 Safety glass 1.6 Appropriate clothing
2. Hand tools	2.1 Adjustable Wrench 2.2 Pipe vise 2.3 Pipe wrench 2.4 Pipe cutter 2.5 C-clamp 2.6 Chisels 2.7 Hack saws 2.8 Die and stock (threader) 2.9 Files 2.10 Hammer 2.11 Measuring Tapes 2.12 Pliers 2.13 Plumb bob 2.14 Punches 2.15 Scrapers 2.16 Screwdrivers 2.17 Taps 2.18 Steel Wire Brush 2.19 Paint Brush 2.20 Oil can 2.21 Sprit level 2.22 Tri-Square 2.23 Vernier Calipers
3. Power tools	3.1 Power drill machine 3.2 Grinding machine 3.3 Threading machine 3.4 Core Cutting Machine 3.5 PPR welding device
4. Tools, equipment and materials	4.1 Tools: 4.2 Measuring tape (5 mtrs.) 4.3 Steel rule 4.4 Caliper 4.5 Adjustable wrench 4.6 Box wrench set

	4.7 Pipe wrench 4.8 Pliers, mechanical 4.9 Die stock 4.10 Pipe cutter 4.11 Hand hacksaw 4.12 Screw driver: Star, flat, positive 4.13 Spirit level 4.14 Pipe reamer 4.15 Files: flat, half round, round 4.16 Equipment 4.17 Pipe threading machine 4.18 Power hacksaw 4.19 Drill press 4.20 Grinder, angle 4.21 Welding machine 4.22 Materials 4.23 Coolant, water based 4.24 Soft stone (Marker) 4.25 Marking pen 4.26 Pencil 4.27 Solvent 4.28 Thread tape (Teflon) 4.29 Welding rods 4.30 Masking tape
5. Pipes	5.1 G.I. pipe 5.2 PVC pipe 5.3 Stainless steel pipe 5.4 B.I. pipe 5.5 Cast iron pipe
6. Deburring tool	6.1 Flat file 6.2 Round file 6.3 Half round file 6.4 Pipe reamer 6.5 Sand paper
7. Coolant	7.1 Water 7.2 Oil 7.3 Air 7.4 Synthetic 7.5 Nitrogen

Curricular Content Guide

1. Underpinning Knowledge	1.1 OSH policies and procedure 1.2 First aid procedure 1.3 Appropriate hand tools and power tools 1.4 Tools, equipment and materials 1.5 Methods of clamping and fixing steel pipes using pipe vice 1.6 Types of pipes 1.7 Procedure of cutting pipes using appropriate cutting tool
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	1.8 De burring tools and their application 1.9 Types of pipe threads 1.10 Purpose of cutting coolants
2. Underpinning Skills	2.1 Measuring and marking pipes 2.2 Clamping and fixing steel pipes using pipe vice 2.3 Cutting pipes using appropriate cutting tool 2.4 Deburring newly cut pipe from burr/sharp edges using appropriate de burring tool 2.5 Cutting threads on pipes in accordance with workplace procedure 2.6 Cleaning workplace and disposing waste materials in accordance with workplace requirement
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Fabric (samples) 4.3 Trims and accessories 4.4 Bill of Materials (samples)

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Measured and marked pipes in accordance to plumbing plan/drawing specification. 1.2 Clamped and fixed Steel pipes using pipe vice. 1.3 Cut pipes using appropriate cutting tool. 1.4 De burred newly cut pipe from burr/sharp edges using appropriate de burring tool. 1.5 Cut threads on pipes in accordance with workplace procedure.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: MAKE PIPE JOINTS	Nominal Duration: 15 Hrs.	Unit Code: SICIP-CON-PLF-2-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to make pipe joints. It specifically includes the tasks of interpreting drawings and plumbing plans, cutting wall and floor, assembling and join pipes and fittings, laying pipes with fittings, cleaning and maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret drawings and plumbing plans	1.1 Building <u>drawing/plumbing plan</u> is gathered. 1.2 Building drawing/plumbing plans are interpreted. 1.3 Work area is identified.
2. Cut wall and floor	2.1 Appropriate place for the pipe access work is marked in accordance with the plumbing plan/ specification 2.2 Concrete wall and floor are cut using core cutting machine to create pipe access in accordance to plumbing plan/specification. 2.3 Walls and floors are grooved without causing damage to the wall or floor and adjacent installations. 2.4 Correct usage of tools and equipment is observed.
3. Assemble and Join Pipes and Fittings	3.1 Tools, equipment and materials are selected, gathered and checked for usability. 3.2 PPEs are checked and used appropriately. 3.3 Pipes are assembled and joined with <u>fittings</u> as required. 3.4 Measurement and alignment are checked
4. Lay pipes with fittings	4.1 Assembled pipes are laid 4.2 Laid pipes are clumped with wall and floor.
5. Clean and maintain the work area, tools & equipment	5.1 Plumbing tools & equipment are cleaned and stored in designated area. 5.2 Work area is cleaned and waste materials are disposed in accordance with workplace requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. Drawing/Plumbing plan	1.1 Plan for water supply system 1.2 Plan for sewerage system 1.3 Plan for Rainwater harvesting system
2. Fittings	2.1 Elbow, 90° 2.2 Elbow. 45°

	2.3 Tee 2.4 Coupling 2.5 Pipe nipple 2.6 Pipe plug
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Curricular Content Guide

1. Underpinning Knowledge	1.1 building drawing/plumbing plans 1.2 Selecting, gathering and checking process of tools, equipment, and materials for usability 1.3 Uses of fittings 1.4 Laying-out process for access work 1.5 Describe cutting of walls and floors without causing damage to the wall or floor and adjacent installation
2. Underpinning Skills	2.1 Interpreting building drawing/plumbing plans 2.2 Selecting, gathering and checking tools, equipment, and materials for usability 2.3 Laying-out for access work 2.4 Cutting walls and floors without causing damage to the wall or floor and adjacent installation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Calculator 4.4 Projector 4.5 Stationery 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Interpreted building drawing/plumbing plans. 1.2 Selected, gathered and checked tools, equipment and materials for usability. 1.3 Laid-out for access work. 1.4 Cut walls and floors without causing damage to the wall or floor and adjacent installations.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration

	2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: INSTALL WATER SUPPLY AND WASTE WATER PIPES	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-PPF-3-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install water supply and wastewater pipes. It specifically includes the tasks of gathering and inspecting tools, equipment and materials, fixing joints and fittings for pipe, installing water supply pipes, installing wastewater pipes and cleaning and maintaining the work area, tools & equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Gather and inspect tools, equipment and materials	1.1 PPE are selected and used. 1.2 Tools, equipment and materials are gathered and checked for usability. 1.3 Pipes are identified and gathered in accordance with plumbing plan specifications.
2. Fix joints and fittings for pipe	2.1 <u>Pipes</u> are measured and marked as per plan/drawing specification 2.2 Pipes are cut within the specified dimension and considering specified tolerance. 2.3 Pipes are made joint and fixed with fittings using appropriate <u>clamping device</u> 2.4 Measurement is checked as per drawing
3. Install water supply pipes	3.1 Pipe runs are positioned on the corresponding location as per plumbing plan. 3.2 Piping joints and fittings are assembled using appropriate tools and <u>sealant</u> . 3.3 Appropriate curing time for sealant after assembly of pipe joints and fittings is observed to create strong bond. 3.4 <u>Clamps and fixtures</u> along pipe runs are installed in accordance with plumbing plan. 3.5 Leaks and <u>non-conformance</u> to plumbing design/plans are checked and corrected.
4. Install wastewater pipes	4.1 Sewer pipes are laid on the trench in accordance with workplace and plumbing plan requirements. 4.2 Bell ended sewer pipes are laid by placing the bell end at the uphill side of the pipe run. 4.3 PVC sewer pipes are installed by using appropriate sealant. 4.4 Soil and vent pipes are placed in accordance with workplace and plumbing plan requirements. 4.5 Alignment and pipe elevation are checked in accordance to workplace and plumbing plan

	requirements.
5. Clean and maintain the work area, tools & equipment	5.1 Plumbing tools & equipment are cleaned and stored in designated area. 5.2 Work area is cleaned and waste materials are disposed in accordance with workplace requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. Pipes	1.1 GI 1.2 PPR 1.3 PVC 1.4 cPVC 1.5 uPVC thread pipe 1.6 uPVC sewer pipe
2. Clamping device	2.1 Pipe stands 2.2 Bench vise w/v-blocks 2.3 Vise grip 2.4 Pipe chuck 2.5 Threading machine chucks
3. Sealants	3.1 Thread tape (Teflon) 3.2 Solvent cement 3.3 cPVC cement 3.4 Plastic-based seals 3.5 Rubber-based seals 3.6 Synthetic seals 3.7 Silicone seals
4. Clamps and fixtures	4.1 Steel pipe hangers 4.2 Plastic pipe hangers 4.3 Pipe clamps (plastic, steel, rubber) 4.4 Tapes
5. Non-conformance	5.1 Leak 5.2 Length (Dimension) 5.3 Pipe size 5.4 Pipe offset angles 5.5 Obstacles 5.6 Aesthetics/appearance

Curricular Content Guide

1. Underpinning Knowledge	1.1 Techniques of measuring and marking pipes 1.2 Procedure and technique of cutting Pipes 1.3 Pipe thread cutting procedure and proper use of thread cutting tools 1.4 Method of positioning and lay outing pipe runs 1.5 Techniques and safety precautions of tightening piping
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	joints and fittings using sealants 1.6 Types of pipes clamps and fixtures and means of Installation 1.7 Procedure in checking and correcting leaks
2. Underpinning Skills	2.1. Measuring and marking pipes are as per plan/drawing specification 2.2. joining and fixing pipes with fittings using appropriate clamping device 2.3. Checking measurement as per drawing 2.4. Assembling piping joints and fittings using appropriate tools and sealant. 2.5. PVC sewer pipes are installed by using appropriate sealant. 2.6. Placing soil and vent pipes in accordance with workplace and plumbing plan requirements
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure (samples) 4.3 Projector 4.4 Stationery 4.5 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified and gathered pipes in accordance with specified sizes/schedules. 1.2 Cut pipes using appropriate cutting tool and within the specified dimension considering specified tolerance. 1.3 Carried out thread cutting on pipes in accordance to workplace procedure. 1.4 Positioned pipe runs at the corresponding areas as per plumbing plan. 1.5 Tightened piping joints and fittings using appropriate tools and sealant. 1.6 Installed clamps and fixtures along pipe runs in accordance with plumbing plan. 1.7 Checked and corrected leaks and non-conformance to
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	plumbing design/plans.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: CARRY OUT PLUMBING FIXTURES INSTALLATION AND TESTING	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-PPF-4-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out plumbing fixtures installation and testing. It specifically includes the tasks of preparing for plumbing fixture installation, gathering tools, equipment and materials, installing plumbing fixtures, performing leak and pressure testing, cleaning and maintaining the work area, tools and equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for plumbing fixture installation	1.1 <u>Work activities</u> are confirmed in accordance with plumbing plan and workplace requirements. 1.2 Work site is visited and prepared prior to installation works. 1.3 <u>Plumbing fixture</u> is determined, collected and checked for workability and quality.
2. Gather tools, equipment and materials	2.1 Required tools, equipment and materials are identified in accordance with plumbing plan 2.2 Plumbing fixtures to be installed are gathered in the work site. 2.3 PPEs are selected and used. 2.4 Tools, equipment and materials are gathered and checked for usability.
3. Install plumbing fixtures	3.1 The installation site is prepared, ensuring a level surface and proper alignment with the drainage outlet or flange. 3.2 Plumbing fixtures are installed in accordance with plumbing plan and following manufacturer's instruction/specification. 3.3 The fixture is positioned correctly, ensuring alignment with the waste outlet or floor fixtures. 3.4 The fixture is secured using appropriate fasteners, ensuring bolts or fixing mechanisms are tightened without over-tightening. 3.5 <u>Sealing material</u> is applied around the base of the fixture according to workplace standards. 3.6 The water supply connections are established securely. 3.7 Check newly installed plumbing fixtures for <u>non-conformance</u> with workplace requirements. 3.8 Reworks/adjustment is carried out in accordance with workplace requirements.
4. Perform leak and pressure testing	4.1 <u>Work activities of testing</u> are identified in accordance with plumbing plan and workplace requirements. 4.2 <u>Pressure testing methods</u> are identified and applied into the system

	4.3 Required <u>tools, equipment and materials for pressure testing</u> are collected. 4.4 Appropriate <u>Leak testing method</u> is identified and applied to determine leaks. 4.5 Leak testing result is recorded and reported.
5. Clean and maintain the work area, tools & equipment	5.3 Plumbing tools & equipment are cleaned and stored in designated area. 5.4 Work area is cleaned and waste materials are disposed in accordance with workplace requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. Work activities	1.1 Western or Asian commode Shower 1.2 Bath tubs 1.3 Wash basin 1.4 Kitchen sink 1.5 Urinal 1.6 Terminal valves/faucets
2. Plumbing fixture	2.1 Western commode 2.2 Asian commode 2.3 Shower 2.4 Bath tubs 2.5 Wash basin 2.6 Kitchen sink 2.7 Urinals 2.8 Geyser 2.9 Hot and cold mixer tap 2.10 Terminal valves/faucets
3. Sealing material	3.1 Concrete/Cement 3.2 Rubber grommet 3.3 Plastic grommet 3.4 White cement 3.5 Wax seal 3.6 Rubber-based seal 3.7 Silicone seal 3.8 Plastic sealants 3.9 Synthetic sealants
4. Non-conformance	4.1 Leaks 4.2 Damaged fixtures 4.3 Broken/damaged fittings 4.4 Wrong Pipe size 4.5 Wrong Fitting size 4.6 Damage flooring/walls
5. Work activities of testing	5.1 Checking tightness of valves/fittings 5.2 Measuring pipe runs 5.3 Identifying piping run which are subject for pressure testing 5.4 Identifying fittings to be isolated 5.5 Isolating system components
6. Pressure testing method	6.1 Pneumatic testing method 6.2 Hydrostatic testing
7. Tools, equipment and	7.1 Tools:

materials for pressure testing	7.1.1 Measuring tape 7.1.2 Adjustable wrench 7.1.3 Box wrench set 7.1.4 Pipe wrench 7.1.5 Pliers, mechanical 7.1.6 Screwdriver: Star, flat, positive 7.1.7 Pressure gauge 7.1.8 Quick connect couplers/adapters 7.2 Equipment 7.2.1 Pressure testing machine 7.2.2 Water pump
8. Leak testing method	8.1 Soap-Sud solution 8.2 Using leak testing materials 8.3 Observing pressure changes

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of plumbing fixtures and their functions 1.2 Procedure of installing plumbing fixture flange 1.3 Tightening techniques plumbing fixture to avoid damage/breakage 1.4 Care in the installation of ceramic type plumbing fixtures 1.5 Methods and procedures of installing different types of plumbing fixtures 1.6 Leak testing and qualifying workmanships in plumbing fixture installation 1.7 Understanding pressure testing methods, procedures, and tools, including allowable pressure levels and equipment accuracy. 1.8 Applying pressure progressively and conducting leak tests at various stages using appropriate methods.
2. Underpinning Skills	2.1 Gathering installed plumbing fixtures in the work site 2.2 Installing plumbing fixture flange securely making sure the bolts are in place 2.3 Placing the new plumbing fixture on the flange aligning the bolt holes with the bolts of the flange and wax ring in its proper place. 2.4 Tightening squarely the nuts with washers without overtightening 2.5 Installing other plumbing fixtures in accordance with plumbing plan and following manufacturer's instruction/specification 2.6 Checking newly installed plumbing fixtures for leaks and non-conformance with workplace requirements 2.7 Determining maximum test pressure and selecting appropriate pressure testing methods based on plumbing plans and workplace requirements. 2.8 Reporting and recording test results while following

	safety precautions and workplace procedures. 2.9 Carrying out reworks /adjustment in accordance with workplace requirements
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Projector 4.4 Stationery 4.5 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Gathered installed plumbing fixtures in work site. 1.2 Installed plumbing fixtures flange securely making sure bolts are in place. 1.3 Placed the new plumbing fixtures on the flange aligning the bolt holes with the bolts of the flange and wax ring in its proper place. 1.4 Checked newly installed plumbing fixtures for leaks and non-conformance with workplace requirements.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.3 Competency assessment must be done at a training center. 3.1 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM INSTALLATION OF HOT WATER PIPING SYSTEM.	Nominal Duration: 10 Hrs.	Unit Code: SICIP-CON-PPF-5-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform installation of hot water piping system. It specifically includes the tasks of preparing for works, installing pipes on the wall surface, installing geyser with service line, cleaning and maintaining the work area, tools & equipment.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for works	1.1 PPE are selected and used. 1.2 Appropriate hand tools and power tools are selected as per job requirement. 1.3 Workplace is selected and prepared as required.
2. Install pipes on the wall surface	2.1 Pipes are measured and marked as per drawing specification. 2.2 Pipes are installed on the wall surface as required 2.3 Pipes are clamped using appropriate <u>clamping materials</u>
3. Install geyser with service line	3.1 Geyser is positioned on the corresponding location as per drawing. 3.2 Pipes and fittings are assembled with geyser 3.3 Leaks and <u>non-conformance</u> to geyser connections are checked and corrected. 3.4 Electric line to geyser is connected and ensured 3.5 Hot water flow from geyser is observed.
4. Clean and maintain the work area, tools & equipment	4.1 Plumbing tools & equipment are cleaned and stored in designated area. 4.2 Work area is cleaned and waste materials are disposed in accordance with workplace requirements.

Range of Variables

Variable	Range (Includes but not limited to):
1. Clamping Materials	1.1 Plastic C-clamp 1.2 Metallic C-clamp 1.3 Pipe hanger 1.4 Pipe sledge 1.5 Tapes
2. Non-conformance	2.1 Leaks 2.2 Length 2.3 Pipe size 2.4 Pipe offset angles 2.5 Obstacles 2.6 Aesthetics/appearance

Curricular Content Guide

1. Underpinning Knowledge	1.1 Methods and procedure of cutting concrete wall and floor to create pipe access 1.2 Pipe and thread cutting procedure and appropriate cutting tools 1.3 Types of pipe threads 1.4 Method of positioning and lay outting steel pipe runs 1.5 Techniques and safety precautions of tightening steel piping joints and fittings using sealants 1.6 Thread sealing techniques
2. Underpinning Skills	2.1 Measuring pipes and marking 2.2 Clamping pipes 2.3 Cutting pipes 2.4 Cutting pipes within the specified dimension and considering specified tolerance 2.5 Positioning pipe runs on the corresponding location as per plumbing plan 2.6 Assembling piping joints and fittings using appropriate tools and sealant 2.7 Installing pipe clamps and fixtures on pipe runs in accordance with plumbing plan 2.8 Checking and correcting leaks and non-conformance to plumbing design/plans 2.9 Cleaning used tools & equipment
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Basic tools of quality (sheets, diagrams, charts) 4.3 Garments (defective samples) 4.4 Work sheets

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Clamped and cut pipes using appropriate clamping device and cutting tool. 1.2 Positioned pipe runs on the corresponding location as
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	per plumbing plan. 1.3 Assembled piping joints and fittings using appropriate tools and sealant. 1.4 Installed pipe clamps and fixtures on pipe runs in accordance with plumbing plan. 1.5 Checking and correcting leaks and non-conformance to plumbing design/plans.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.4 Competency assessment must be done at a training center. 3.1 Assessment should be done by the certified assessor of TÜV SÜD.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Plumbing and Pipe Fitting
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
 - Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm) Space for Plumbing booth not included

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops with internet facility	1
2.	Multimedia projector & screen	1
3.	Plumbing workstation/ booth with fitting fixture installation facility	5
4.	Kitchen Sink	2
5.	Wash Basin	5
6.	Western commode	5
7.	Asian commode	5
8.	Urinal	2
9.	Shower unit	5
10.	Bath tub	1
11.	Geyser	2
12.	Hot and cold-water mixer tap	5
13.	Pipe Cutter	20
14.	Pipe vice (with stand/ work table)	20
15.	Pipe threading machine	1
16.	Diestock	20
17.	Grinding Machine	10
18.	Spirit level	10
19.	Drill machine	5
20.	PPR welding machine	2
21.	Core cutting machine	1
22.	Pipe wrench	20
23.	Overhead water Tank	1
24.	Tri-Square	10
25.	Plum bob	10
26.	Measuring Tape	20

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Plumbing and Pipe Fitting** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.